

THE MONOMIAL $x_0x_1^4x_2^5$ IS NOT 2-COMPUTABLE UNDER DEFINITION 3.5 AS PRINTED

ABSTRACT. Carlini, Catalisano, Chiantini, Geramita and Woo introduced e -computability as a tool for Strassen's additivity conjecture, and asked in their Remark 6.2 whether $G_1 = x_0x_1^4x_2^5$ is 2-computable. Under their Definition 3.5 as printed, the answer is no. The reason is a one-line upper bound: if $F \in S_d$ is e -computable then $e \cdot \text{rk}(F) \leq N(F) - \sum_{j < e} \text{HF}(T/F^\perp, j)$, where $N(F) = \dim_k T/F^\perp$; for G_1 this reads $60 \leq 56$. A remark records that the same bound forces $e \leq a_0$ for every monomial $x_0^{a_0} \cdots x_n^{a_n}$ with $0 < a_0 \leq \cdots \leq a_n$. Nothing here bears on Strassen's conjecture itself, and nothing here covers readings of e -computability that drop the clause "generated in degree e ".

1. THE QUESTION

Let k be algebraically closed of characteristic 0, let $S = k[x_0, \dots, x_n]$ and $T = k[X_0, \dots, X_n]$, and let T act on S by $X_i \mapsto \partial/\partial x_i$. For $0 \neq F \in S_d$ put $F^\perp = \text{Ann}(F) \subset T$ and $A_F = T/F^\perp$; this is an artinian Gorenstein algebra of socle degree d , so

$$\text{HF}(A_F, j) = \text{HF}(A_F, d - j), \quad \text{HF}(A_F, j) > 0 \iff 0 \leq j \leq d,$$

and we write $N(F) = \dim_k A_F = \sum_j \text{HF}(A_F, j)$. Finally $\text{rk}(F)$ is the Waring rank of F .

Carlini, Catalisano, Chiantini, Geramita and Woo [1] introduced the following notion.

Definition 3.5 of the source. $F \in S_d$ is e -computable if there exists an ideal $I \subset T$ generated in degree e such that

$$(1) \quad \text{rk}(F) = \frac{1}{e} \sum_i \text{HF}(T/(F^\perp : I + (t)), i)$$

for general $t \in I_e$.

That is their definition restated in the notation fixed above; verbatim, Definition 3.5 of [1] reads "there exists an ideal $I \subset T$ generated in degree e such that for general $t \in I_e$ we have $\text{rk}(F) = (\frac{1}{e}) \sum_{i=0}^\infty \text{HF}(T/(F^\perp : I + (t)), i)$ ". The clause "generated in degree e " is load-bearing for everything below; see the Scope paragraph. Their Remark 6.2, on journal page 387, contains the question we answer:

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If $G_1 = x_0x_1^4x_2^5$ then we showed that G_1 is 1-computable and $\text{rk}(G_1) = 30$. But we do not know if G_1 is 2-computable. Thus we cannot use the theorem to find the rank of $F + G_1$, although Strassen's conjecture says that the rank should be $25 + 30$. However, if $G_2 = x_0^3x_1^4x_2^5$, by Proposition 4.2, we know that G_2 is 2-computable and $\text{rk}(G_2) = 30$. Hence $\text{rk}(F + G_2) = 25 + 30 = 55$.

Proposition 4.2 of [1] makes $x_0^{a_0} \cdots x_n^{a_n}$, with $0 < a_0 \leq \cdots \leq a_n$, e -computable for $1 \leq e \leq (a_0 + 1)/2$; for G_1 that gives only $e = 1$. Neither that proposition nor Corollary 3.4 of [1] bounds e from above, and their Remark 4.3 asks whether such monomials are “ e -computable for e 's different from those described in the two propositions”.

Theorem 1. $G_1 = x_0x_1^4x_2^5$ is not 2-computable in the sense of Definition 3.5 of [1] as quoted above. Consequently the question of Remark 6.2 of [1] has a negative answer.

2. THE OBSTRUCTION

Everything follows from one containment. Write $T_{\geq m} = \bigoplus_{i \geq m} T_i$, and read an empty sum as 0.

Lemma 2. Let $0 \neq F \in S_d$, let $e \geq 1$, let $I \subset T$ be any ideal generated in degree e , and let $t \in T$ be any homogeneous element. Then $T_{\geq d-e+1} \subseteq F^\perp : I$, and

$$\begin{aligned}
 (2) \quad \sum_i \text{HF}(T/(F^\perp : I + (t)), i) &\leq \sum_{i=0}^{d-e} \text{HF}(A_F, i) \\
 &= N(F) - \sum_{j=0}^{e-1} \text{HF}(A_F, j) < N(F).
 \end{aligned}$$

Proof. Step 1. Since $I = (I_e)$, it suffices for $T_{\geq d-e+1} \subseteq F^\perp : I$ to check that $hg \in F^\perp$ whenever $\deg h \geq d - e + 1$ and $g \in I_e$: then $\deg(hg) \geq d + 1$, and every form of degree $> d$ annihilates F .

Step 2. The containment is strict over $F^\perp$ when $1 \leq e \leq d + 1$, because $\text{HF}(A_F, d - e + 1) > 0$ forces $T_{d-e+1} \not\subseteq F^\perp$.

Step 3. Put $J = F^\perp : I + (t)$; since $F^\perp$, I and t are homogeneous, so is J , and T/J is graded, so its Hilbert function is defined. By Step 1, $F^\perp + T_{\geq d-e+1} \subseteq F^\perp : I \subseteq J$, and all three quotients are artinian, so each Hilbert-function sum is a k -dimension and

$$\begin{aligned}
 \sum_i \text{HF}(T/J, i) &= \dim_k T/J \leq \dim_k T/(F^\perp : I) \leq \dim_k T/(F^\perp + T_{\geq d-e+1}) \\
 &= \sum_{i=0}^{d-e} \text{HF}(A_F, i).
 \end{aligned}$$

The equality in (2) is Gorenstein symmetry, and the last strict inequality holds because $\text{HF}(A_F, 0) = 1$. $\square$

Beyond homogeneity, nothing in the proof uses any property of t ; in (1) one has $t \in I_e$, so t is homogeneous of degree e . Hence (2) holds under either reading of “for general $t \in I_e$ ” — a generic t , or merely some t — and the following corollary is insensitive to that choice.

Corollary 3. *If F is e -computable then $e \cdot \text{rk}(F) \leq N(F) - \sum_{j=0}^{e-1} \text{HF}(A_F, j) < N(F)$.*

Proof of Theorem 1. Here $d = 10$ and $G_1^\perp = (X_0^2, X_1^5, X_2^6)$, the apolar ideal of a monomial: a monomial $X_0^{c_0} X_1^{c_1} X_2^{c_2}$ kills $x_0 x_1^4 x_2^5$ exactly when $c_0 > 1$ or $c_1 > 4$ or $c_2 > 5$. It is a monomial complete intersection, so A_{G_1} has

$$(\text{HF}(A_{G_1}, j))_{j=0}^{10} = (1, 3, 5, 7, 9, 10, 9, 7, 5, 3, 1), \quad N(G_1) = 2 \cdot 5 \cdot 6 = 60.$$

By the monomial Waring-rank theorem of Carlini, Catalisano and Geramita [2], $\text{rk}(G_1) = (4+1)(5+1) = 30$, the same value [1] prints on page 387. If G_1 were 2-computable, Corollary 3 at $e = 2$ would give

$$60 = 2 \cdot 30 = e \cdot \text{rk}(G_1) \leq N(G_1) - \text{HF}(A_{G_1}, 0) - \text{HF}(A_{G_1}, 1) = 60 - 1 - 3 = 56,$$

which is false, with margin 4. Hence G_1 is not 2-computable. $\square$

Remark. The same bound also caps the right-hand side of (1) for G_1 at $e = 2$ by 56, so the best lower bound for $\text{rk}(G_1)$ obtainable at $e = 2$ from Corollary 3.4 of [1] is $\lfloor 56/2 \rfloor = 28 < 30$. Applied to $F = x_0^{a_0} \cdots x_n^{a_n}$ with $0 < a_0 \leq \cdots \leq a_n$, for which $F^\perp = (X_0^{a_0+1}, \dots, X_n^{a_n+1})$, $N(F) = \prod_i (a_i + 1)$ and $\text{rk}(F) = N(F)/(a_0 + 1)$ by [2], Corollary 3 likewise forces $e \leq a_0$ whenever F is e -computable: for $e \leq d + 1$ it gives $e(\text{rk}(F) + 1) \leq (a_0 + 1)\text{rk}(F)$, hence $e < a_0 + 1$, and for $e > d + 1$ the left side of (2) is the empty sum $0 < e \cdot \text{rk}(F)$. No converse is claimed.

3. VERIFICATION

Every number above is a small exact count, and the proof of Theorem 1 can be redone by hand from the exponent vector $(1, 4, 5)$. The companion program `verify.py` (Python, standard library only, exact integer and `Fraction` arithmetic) re-derives it mechanically: it recomputes $G_1^\perp$ from the apolarity pairing and confirms it equals the printed ideal on every monomial of degree ≤ 13 ; obtains $\text{HF}(A_{G_1})$ twice, by counting standard monomials and by ranks of the apolarity matrices; checks the three steps of Lemma 2 at $e = 2$; and evaluates the cap 56 as a dimension. It quotes $\text{rk}(G_1) = 30$ from [2] rather than reproving it. The transcript also records finite censuses and control computations that bear on statements no longer made here; those are finite checks and establish nothing universal. Lemma 2 itself, which is a statement about arbitrary F , e , I and t , is proved by hand above and is not established by any finite computation. All 55 of its checks pass; the recorded transcript is `verify.output.txt`.

Scope. The clause “generated in degree e ” in Definition 3.5 is what makes Step 1 of Lemma 2 work. If I is allowed to contain forms of degree $< e$, Step 1 fails and nothing here settles Remark 6.2; restatements of the notion that do not carry that clause are in circulation, in the authors’ withdrawn preprint arXiv:1502.01107, in the survey [4], and in [3], where it appears instead as a hypothesis of Theorem 2.2. So what is settled is Theorem 1 under Definition 3.5 as printed in [1]. Lemma 2 and Corollary 3 are characteristic-free, but $\text{rk}(G_1) = 30$ is a characteristic-zero theorem [2], so Theorem 1 is stated over a field of characteristic 0. The bound is necessary, not sufficient, so it settles nothing where it does not fire; in particular it says nothing about the negative $e = 1$ results of [1]. Nothing here determines $\text{rk}(F + G_1)$ or bears on Strassen’s additivity conjecture: Theorem 1 closes off the route of Remark 6.2 rather than completing it, and the value $\text{rk}(F + G_2) = 25 + 30 = 55$ in the passage quoted above is the assertion of [1], not of this note. Statement numbers are those of the journal version of [1].

Attribution and adjacent work. The shape of the argument is the source’s own: Remark 7.3 of [1] already bounds the same Hilbert-function sum from above in order to defeat 1-computability of $w(x^3 + y^3 + z^3)$. Corollary 3.4 of [1] uses the sum in (1) to bound $\text{rk}(F)$ from below; Lemma 2 bounds that sum from above by an invariant of F alone, uniformly in I and in homogeneous t .

REFERENCES

- [1] E. Carlini, M. V. Catalisano, L. Chiantini, A. V. Geramita, and Y. Woo, *Symmetric tensors: rank, Strassen’s conjecture and e -computability*, Ann. Sc. Norm. Super. Pisa Cl. Sci. (5) **XVIII** (2018), no. 1, 363–390. doi:10.2422/2036-2145.201602_004; preprint arXiv:1506.03176v2.
- [2] E. Carlini, M. V. Catalisano, and A. V. Geramita, *The solution to the Waring problem for monomials and the sum of coprime monomials*, J. Algebra **370** (2012), 5–14. doi:10.1016/j.jalgebra.2012.07.028.
- [3] M. Bhat, E. Carlini, S. Dubey, and S. K. Masuti, *Waring decompositions of the product of two quadrics: the small rank cases*, Linear Algebra Appl., doi:10.1016/j.laa.2026.06.027; arXiv:2511.23035.
- [4] A. Bernardi, E. Carlini, M. V. Catalisano, A. Gimigliano, and A. Oneto, *The Hitchhiker guide to: secant varieties and tensor decomposition*, Mathematics **6** (2018), no. 12, art. 314. doi:10.3390/math6120314; arXiv:1812.10267.